

LS100 Module 3: Computational Behavioral Sciences

Curriculum & Weekly Timeline

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Task	Timeline
Task: Drafting research proposal and research team. Reading: Guide01: Writing Research Plans and Proposals Deliverable: Submit Project Proposal in 200-300 words. Frame research questions, why this is important, hypotheses, test/ experimental approach, expected outcome.	week 00
Deliverable: Presentation 01 - Work Plan - Storytelling format: Background → Hypothesis → Experiment Design → Tool Design (optional) → Data collection → Data Analysis → Results → Report. Reading: - Guide01: Writing Research Plans and Proposals - Guide02: Getting Started with Python Lab: Notebook01: Introduction to Python programming - Data in Python	Week 01
Task: Experimental Design and Data Schema - Primary “raw” data format - image/ video/ audio/ digital records - Defining data schema: variables, metrics; Create data labeling guideline - Define data collection methods, success criteria; Schedule data collection. Lab: Notebook02: Data Automation Deliverable: Build a minimal protocol with data management plan (DMP).	Week 02
Task: Preliminary Data collection and Redesign Work Plan - Recording fieldcraft: - Video basics: camera placement, fps, shutter speed, aperture.	Week 03

● Audio basics: Signal vs noise, room treatment. ● Data collection: Using AI for data collection - Computer vision and Audio models ● Collect and label preliminary data ● Redesign the experiment with updated understanding of data ● Introduction to GitHub Lab: ● Notebook03: Data Visualization. ● Audio Basics/ Video Basics	
Task: Data Processing ● Data processing, convert primary data into analyzable format – secondary data ● Finalizing methods of converting secondary data to tertiary dataset. Lab: Notebook04: Computational Thinking & Fluency - Data Structure and Algorithms (DSA) Optional Lab: Notebook05: Introduction to Python Classes and Decorators Deliverable: ● Notebook: converting primary data to secondary data ● Notebook: Visualizing your secondary data.	Week 04
Task: Data Processing ● Data Collection and labeling ● Creating tertiary dataset, e.g., ○ Feature clusters or segments ○ Kinematics (velocity / angles/ counts). Reading: Art of Communicating Science Deliverable: ● Notebook: Converting secondary data to tertiary datasets. ● Document: First draft of project report.	Week 05
Deliverable: Presentation 02 - Work Progress ● Research Question; Hypothesis; Experimental Design; Data schema; Data collection status; Data visualization from primary, secondary, and tertiary data; Projected timeline. Reading: Art of Communicating Science	Week 06

Task: Introduction to Statistical Tests • Data Collection and labeling • Data Wrangling • Choosing Appropriate Statistical Tests Deliverable: Registered Analysis Plan (why you chose these tests, explain model formulas, strategies for predictive tasks).	Week 07
Task: Data Finalization • Data collection and labeling ends. • Data Wrangling Deliverable: Data Documentation linked to data uploaded on cloud.	Week 08
Task: Data Analysis • Data Analysis Sprint • Create first graph / plot for final report. Lab: Conducting Data Analyses Optional Reading: Using AI to Make Sense of Data: When to Use ML (vs. classical stats)?	Week 09
Task: Data Analysis & Report Writing • Analysis Sprint + Writing Window • Create graphs/ plots for final report. • Code + data packaging; Documentation; README, quick start, config file.	Week 10
Task: Data Analysis & Report Writing • Analysis Sprint + Writing Window • Writing focus: visualization / storytelling.	Week 11
Task: Report Writing & Preparing for Final Presentation • Writing focus: Polishing. Review final report. • Practice group final presentation. • Peer review: Structured rubric (clarity, methods, results, limitations, ethics, reproducibility). Deliverable: Submission of the first draft of the final report.	Week 12

Deliverable: Presentation 03 - Final Presentation

- Research Question; Hypothesis; Experimental Design; Data visualization; Research Findings; Conclusion; Future Direction.

Week 13

Task: Incorporate feedback to the final report.

Deliverable: Submit final report.

Week 14